

JEE MAIN REVISION MASTER: GASEOUS STATE OUT OF SYLLABUS

Subject: Chemistry | Target Scope: JEE Main Previous Years | Empirical Base: 63 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Gaseous State - Core Application	63 PYQs	General Concept

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
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3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Gaseous State - Core Application (Very High Yield — 63 PYQs)

Core Principle: Key problem pattern in Gaseous State.

■ ■ Caution / JEE Pitfall: Watch for algebraic and sign errors.

■ Speed Trick: Verify dimensions and boundary conditions.

Representative Source PYQs:

- [2025 Morning Shift] Q#1 (MCQ): "At the sea level, the dry air mass percentage composition is given as nitrogen gas: 70.0, oxygen gas: 27.0 and argon gas: 3.0. If total pr..." [Ans: N/A]
- [2023 Morning Shift] Q#1 (Numerical): "Three bulbs are filled with CH_4 , CO_2 and Ne as shown in the picture. The bulbs are connected through pipes of zero volume. When the..." [Ans: 3]
- [2023 Evening Shift] Q#2 (MCQ): "Arrange the following gases in increasing order of van der Waals constant 'a' A. Ar B. CH_4 C. H_2O D. C_6H_6 Choose the corre..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2025 (Online) 7th April Morning Shift (Q#1)

At the sea level, the dry air mass percentage composition is given as nitrogen gas: 70.0 , oxygen gas: 27.0 and argon gas: 3.0 . If total pressure is 1.15 atm , then calculate the ratio of following respectively: (i) partial pressure of nitrogen gas to partial pressure of oxygen gas (ii) partial pressure of oxygen gas to partial pressure of argon gas (Given: Molar mass of N, O and Ar are 14, 16 and 40gmol⁻¹ respectively.)

- (A) 5.46,17.8
- (B) 2.96,11.2
- (C) 4.26,19.3
- (D) 2.59,11.85

Correct Answer: None

Solution Analysis: Partial Pressure Ratio of Nitrogen to Oxygen : The partial pressure of a gas is calculated using its mole fraction. The mole fraction is the ratio of the number of moles of the gas to the total number of moles of all gases. For nitrogen (N₂), the mole fraction $\frac{n_{\text{N}_2}}{n_{\text{N}_2} + n_{\text{O}_2} + n_{\text{Ar}}}$ can be simplified as $(70/28 / 70/28 + 27/32 + 3/40)$. For oxygen (O₂), the mole fraction is $(27/32 / 70/28 + 27/32 + 3/40)$. Therefore, the ratio of the partial pressure of nitrogen to oxygen is : $\frac{P_{\text{N}_2}}{P_{\text{O}_2}}$